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Clinical Characteristics of UM and Association of Metastasis of Uveal Melanoma with Congenital Oculocutaneous Melanosis in Asian Patients: Analysis of 1151 Consecutive Eyes

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Purpose: To analyze the clinical characteristics of uveal melanoma (UM) and evaluate the relationship of congenital oculocutaneous melanosis (OCM) to the prognosis of Asian patients with UM.

Design: Retrospective cohort study.

Participants: We included a total of 1151 Asian patients with UM who were managed at the Beijing Tongren Hospital from June 26, 2005, to July 27, 2020.

Methods: I-125 plaque brachytherapy, local resection, thermotherapy, or enucleation.

Main Outcome Measures: Melanoma-related metastasis and death.

Results: Of 1151 Asian patients with UM, congenital OCM was present in 23 (0.20%). The melanocytosis involved the conjunctiva (78%), sclera (74%), eyelid (70%), face (26%), forehead (2.2%), iris (0.87%), choroid (0.87%), and auricle (0.4%). Univariate analysis of Cox proportional hazards regression model showed that age, tumor thickness, largest tumor basal diameter, and ciliary body involvement were the risk factors for the poor prognosis of Asian patients with UM. By multivariable analysis, the only factor predictive of melanoma-related metastasis and death was the largest tumor basal diameter (hazard ratio [HR], 1.21 [1.11–1.33], $P < .001$; 1.17 [1.01–1.35], $P = 0.033$). Probability-of-censoring weighted (IPW) estimation was used to mitigate selection bias due to loss to follow-up. Probability-of-censoring weighted estimation revealed the largest tumor basal diameter and ciliary body involvement were associated with metastasis (HR, 1.29 [1.15–1.45], $P < 0.001$; HR, 2.64 [1.01–6.90], $P < 0.048$). During the median follow-up period of 53 (33–67) months, 2 patients with OCM (8.7%) developed melanoma-related metastasis. By using nested case-control design and matched with the factors of age, largest tumor basal diameter, tumor thickness, and ciliary body involvement, Kaplan–Meier survival analysis showed that UM combined with OCM did not increase the risk of melanoma-related metastasis and death ($P = 0.68$, log-rank test).

Conclusions: The prominent risk for metastasis from UM was the largest tumor basal diameter in Asian patients. Estimation of IPW revealed that the largest tumor basal diameter and ciliary body involvement were the risk factors for UM metastasis. Patients with UM combined with OCM had a similar risk for metastasis compared with those with no OCM. *Ophthalmology Retina 2021;5:1164-1172 © 2021 by the American Academy of Ophthalmology*

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Uveal melanoma (UM) is the most common primary intraocular malignancy in adults. The age-standardized incidence rate was approximately 5 to 7 per million population in the White population,[1,2] whereas the incidence rate in the Asian population was approximately 0.3 to 0.6 cases per million population.[3-5] In addition to a relatively lower prevalence, clinical characteristic discrepancies also exist between Asian and White populations. Asian patients with UM tend to be younger at diagnosis and to have a better prognosis.[6,7] Moreover, the association between the tumor thickness and the poor prognosis seems not to exist in Asian patients.[8]

Oculocutaneous melanosis (OCM) was a rare congenital condition characterized by hyperpigmentation of periocular cutaneous involvement, which was called "oculo(dermal) melanocytosis" in 1939 (nevus of Ota).[9,10] In most cases, the dark pigmentation was unilateral, and the melanocytic pigmentation can involve the dermal, ocular alone, or a combination of the 2 conditions. Oculocutaneous melanosis was rare in the general population. Gonder et al[11] estimated that OCM affected only approximately 0.04% of the White

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population and approximately 1.4% of UM with ocular melanocytosis. Cowan and Balistocky[12] reviewed 25 000 patients from eye clinics in 5 years and revealed only 4 cases of ocular melanocytosis. They estimated that the incidence was 1 in 6200 in the White and Black combined population.[12] Oculocutaneous melanosis seemed to be more common in the Asian population;[13] however, the detailed data were still undetermined.

In the published literature, the main concern with OCM was an increased risk for the development of UM.[12-24] Previous studies have been described as an association between OCM and UM in the White population.[12-24] Singh et al[17] estimated the lifetime prevalence of UM in White patients with OCM to be 2.6 cases per 1000 population, compared with an estimated prevalence of UM of 7.5 cases per 10 000 in the general population. The 10-year rate of metastasis was 48% for those with OCM compared with 24% for those without OCM.[23,24] However, Gonder et al[14] reported 17 patients with oculo(dermal) melanocytosis among 1250 White patients with UM and noted that the melanoma was similar in size, cell type, and metastatic risk compared with those with no melanocytosis.

Although the prevalence of and the association of OCM in the White population with UM were known, the prevalence of UM and a relevant estimation in the Asian population were still unknown. To further explore the relationship with UM, in this study, we have analyzed clinical characteristics of 1151 Asian patients with UM and estimated the influence of OCM on patient risk for melanoma-related metastasis and death in the Asian population.

Methods

Participants

A retrospective cohort study consisted of 1151 Chinese patients with a clinical diagnosis of UM treated at the Beijing Tongren Eye Center, Beijing Tongren Hospital, from June 26, 2005, to July 27, 2020. The study and data collection were compliant with the principles of the Declaration of Helsinki. The study was approved by the Medical Ethics Committee of the Beijing Tongren Hospital, and written informed consent was obtained from all participants. All patients with UM enrolled in this study recorded and obtained detailed information regarding the presence and extent of OCM. All patients were examined by the senior ophthalmologist (W.W.).

The demographic data included patient age at diagnosis (years), sex, and race/ethnicity (Asian, Chinese). The following clinical data were included: best-corrected visual acuity, laterality, intraocular pressure, ocular or oculodermal melanocytosis (present or absent), location of melanocytosis (forehead to the hairline or scalp, auricle, face, eyelid, sclera, conjunctiva, iris, or choroid), laterality of melanocytosis with UM (ipsilateral or contralateral), original location of tumor (iris, ciliary body, or choroid), quadrant location of tumor (superior, temporal, inferior, nasal, or macula), tumor configuration (mushroom, dome, plateau, or lentiform), pigmented (included dark black, brown, or mixed) or

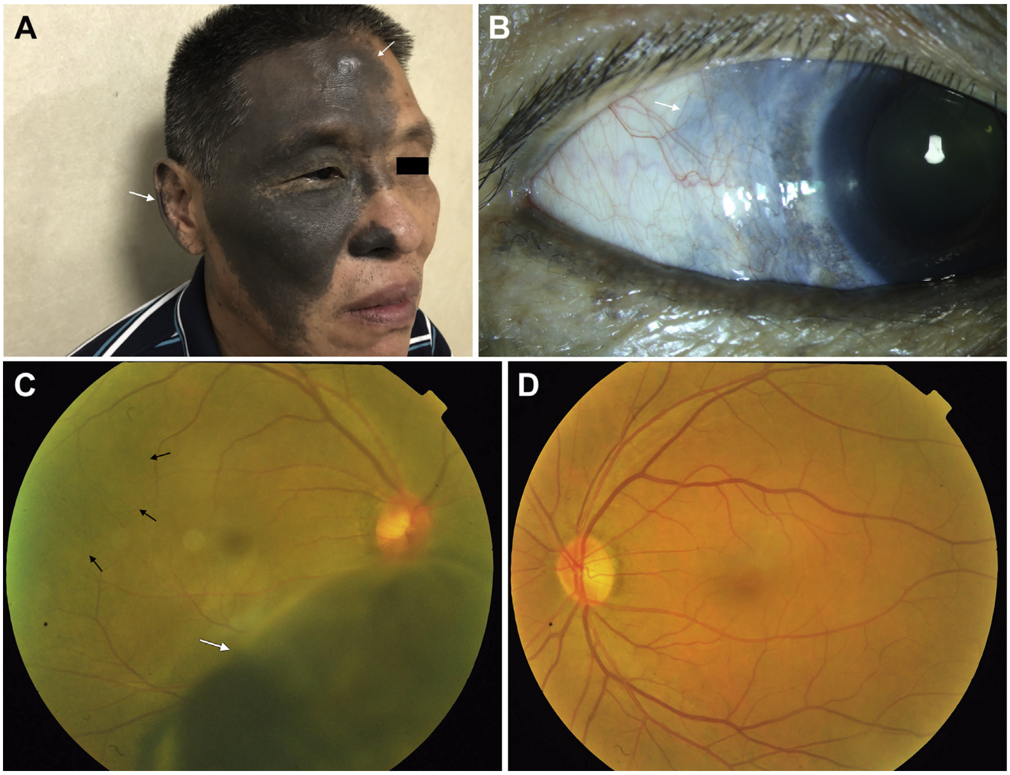

Figure 1. **A,** A 55-year-old Asian man with uveal melanoma (UM) combined with oculocutaneous melanosis (OCM) who had the skin pigmentation in his right forehead, ear, cheeks, nose, eyelid, and skin folds (**white arrow**). **B,** Right eye with black conjunctival and episcleral pigmentation (**white arrow**). **C,** Fundus examination revealed a large pigmented mushroom-shaped choroidal melanoma with peripheral exudative retinal detachment (**white arrow**). The right fundus was dark (**dark arrows**) in color in comparison with the normal left eye **(D)**. **D,** The normal left fundus is in orange color.

nonpigmented, Bruch membrane rupture, exudative retinal detachment, vitreous hemorrhage, and extraocular extension. Largest tumor basal diameter and tumor thickness were measured by standard ocular color Doppler ultrasonography imaging. All findings were documented appropriately with anterior or posterior segment photography, fluorescein angiography and indocyanine green angiography, color Doppler ultrasonography imaging, and magnetic resonance imaging.

Management of UM was performed by standard methods (I-125 plaque brachytherapy, local resection, thermotherapy, or enucleation) following informed consent. The majority of patients with UM received iodine-125 brachytherapy of eye-conserving therapy (n = 929). Enucleation was indicated for advanced UM (large volume at baseline), extraocular extension, or secondary neovascular glaucoma combined with poor visual acuity (n = 165). Local resection is only indicated for resectable iris UM or diagnostic confirmation (n = 57). After ocular treatment of UM, schedule monitoring per 1 month during months 1 to 3, per 3 months during months 4 to 24, per 6 months during months 25 to 60, and per 1 year during the month 60∼ period was provided to the senior ophthalmologist (W.W.). During each follow-up visit, ocular examination (color photography, ultrasonography) and systematic examination (pulmonary computed tomography and abdominal ultrasonography) were performed. Metastasis and metastasis-related death were identified by medical records and confirmed with pathology or autopsy. Metastasis was only confirmed according to pathological diagnosis from the biopsy. Specific death information included the date and cause of death.

Nested Case-Control Study

Oculocutaneous melanosis was defined as congenital melanocytic hyperpigmentation affecting more than 1 area of ocular or periocular tissue (forehead to the hairline or scalp, auricle, face, eyelid, sclera, conjunctiva, iris, or choroid), and location of melanocytosis was recorded (Fig 1). To eliminate confounding factors, we matched with 56 patients with UM without OCM (no melanocytosis group). Patients were selected from 1128 patients with UM without OCM and matched for age (within 5 years), tumor thickness (within 1 mm), tumor basal diameter (within 2 mm), and ciliary body involvement (Tables 1−4). If more than 4 controls were eligible, we randomly chose 4 controls by random number table. A comparison of survival distribution according to the presence or absence of OCM was analyzed by log-rank test.

Table 1. Clinical Characteristics of Asian Patients with UM

	UM Received Plaque Radiotherapy (n = 929)	UM Received Enucleation (n = 165)	UM Received Local Resection (n = 57)
Age, yrs	47.3 (12.4)	46.8 (13.2)	39.3 (13.8) *
Female (%)	465 (50.1%)	77 (46.7%)	26 (45.6%)
Right eye (%)	465 (50.1%)	73 (44.2%)	22 (38.6%)
BCVA (n = 979)			
<0.1	261	35	7
0.1−0.24	201	7	1
0.25−0.49	141	1	2
0.5−0.79	150	3	1
≥0.8	162	6	1
IOP (n = 973)	13.0 [11.2−15.4]	13.6 [11.5−16]	15.0 [11.8−15.8]
Tumor origin (n = 1149)			
Uvea	911 (98.2%)	133 (81.1%)*	29 (50.9%)*
Ciliary body	14 (1.5%)	30 (18.3%)*	24 (42.1)*
Iris	3 (0.3%)	1 (0.6%)*	4 (7.0%)*
Largest tumor basal mm (n = 1017)	11.6 [10.1−13.5]	14.3 [11.1−19.0]*	11 [10.1−14.9]
Tumor thickness mm (n = 1018)	6.6 [4.7−8.5]	10.9 [8.1−13.4]*	8.1 [6.0−10.3]*
Retinal detachment (n = 1145)	664 (71.5%)	123 (68.5%)	16 (28.1%)*
Tumor location (n = 1053)			
Superior	58	6	5
Superior temporal	228	4	7
Temporal	164	16	8
Inferior temporal	187	12	6
Inferior	62	6	7
Inferior nasal	82	1	8
Nasal	68	20	7
Superior nasal	67	9	5
Macular	10	0	0
Optic disc involved (n = 1149)	24 (2.6%)	258 (17.1%)*	0 (%)
Ciliary body involved (n = 1148)	66 (7.1%)	46 (28.2%)*	29 (50.9%)*
Intraocular hemorrhage (n = 1145)	53 (5.7%)	12 (7.3%)	2 (3.5%)
Median follow-up duration (n = 1056)	25.2 [11.2−48.0]	25.4 [13.7−37.7]	59.4 [33.5−77.1]*
UM metastasis†	60 (6.5%)	15 (9.1)	5 (8.8%)
Metastasis-related Death†	22 (2.4%)	12 (13.6%)	6 (15.8%)

Data were weighted estimates and expressed as mean (standard error) or median [percentile 25 − percentile 75] when appropriate.
BCVA = best-corrected visual acuity; IOP = intraocular pressure; UM = uveal melanoma.
*$P < 0.05$ compared with patients with UM who received plaque radiotherapy.
†Excluded the patients of follow duration = 0.

Table 2. Survival Analysis of Patients with UM

	UM Metastasis		Metastasis-related Death		IPW UM Metastasis		IPW Metastasis-related Death	
Univariable Analysis	*HR (95% CI)*	*P Value*	*HR (95% CI)*	*P Value*	*HR (95% CI)*	*P Value*	*HR (95% CI)*	*P Value*
Age (per 10-yr increase)	1.17 [0.99−1.39]	0.062	1.17 [0.92−1.50]	0.20	1.24 [0.99−1.57]	0.064	1.41 [0.95−2.10]	0.092
Sex (male vs. female)	1.25 [0.82-1.92]	0.30	1.09 [0.59−2.03]	0.78	1.36 [0.78−2.40]	0.28	1.32 [0.52−3.36]	0.56
Largest tumor basal	1.23 [1.13−1.34]	**<0.001**	1.22 [1.07−1.40]	**0.003**	1.31 [1.19−1.46]	**<0.001**	1.29 [1.16−1.44]	**<0.001**
Tumor thickness	1.09 [0.99−1.20]	0.079	1.20 [1.04−1.39]	**0.014**	1.12 [0.98−1.28]	0.086	1.34 [1.08−1.67]	**0.009**
Retinal Detachment	1.12 [0.70−1.79]	0.63	0.84 [0.44−1.61]	0.60	1.47 [0.75−2.87]	0.26	0.78 [0.28−2.18]	0.63
Optic disc involved	0.48 [0.07−3.46]	0.47	NA	NA	NA	NA	NA	NA
Ciliary body involved	2.77 [1.49−5.16]	**0.001**	4.03 [1.76−9.25]	**0.001**	4.23 [1.65−10.8]	**0.003**	4.87 [1.43−16.5]	**0.011**
Intraocular hemorrhage	0.48 [0.12−1.99]	0.31	NA	NA	0.90 [0.21−3.81]	0.89	NA	NA
Pathology: mix vs. spindle	1.99 [0.57−6.93]	0.28	1.73 [0.39−7.66]	0.47	1.46 [0.53−4.01]	0.46	1.38 [0.47−4.08]	0.55
Pathology: epithelioid vs. spindle	2.22 [0.51−9.62]	0.29	1.27 [0.17−9.74]	0.81	NA	NA	NA	NA
Multivariable analysis								
Age (per 10-yr increase)	1.17 [0.96−1.43]	0.11	1.24 [0.91−1.67]	0.17	1.28 [0.99−1.66]	0.058	1.35 [0.86−2.12]	0.20
Largest tumor basal	1.21 [1.11−1.33]	**<0.001**	1.17 [1.01−1.35]	**0.033**	1.29 [1.15−1.45]	**<0.001**	1.23 [1.04−1.45]	**0.018**
Tumor thickness	1.01 [0.91−1.13]	0.80	1.13 [0.97−1.33]	0.11	0.99 [0.85−1.15]	0.85	1.19 [0.98−1.45]	**0.084**
Ciliary body involved	1.57 [0.64−3.84]	0.32	1.92 [0.53−7.05]	0.32	2.64 [1.01−6.90]	**0.048**	2.12 [0.77−5.85]	0.15

CI = confidence interval; HR = hazard ratio; IPW = inverse probability-of-censoring weighted; NA = not available; UM = uveal melanoma.
Bolding indicates significant value.

Statistical Analysis

The demographics and tumor characteristics of patients with UM were summarized according to the presence or absence of OCM. Data collected on a continuous scale, including age (years), largest tumor basal diameter (millimeters), and tumor thickness (millimeters), were expressed as mean, median, minimum, and maximum, and they were evaluated with the Student *t* test. Wilcoxon matched-pairs signed-ranks test and chi-square test were used when appropriate for determining differences in cross-sectional characteristics.

We used inverse probability-of-censoring weighted (IPW) estimation to mitigate selection bias due to loss to follow-up.[25,26] In brief, we defined cases with <36 months of follow-up and without the outcome event (metastasis or death) as short follow-up cases and cases ≥36 months of follow-up or with the outcome event (metastasis or death) as long follow-up cases. Logistic regression was used to calculate the possibility (Pr) of short follow-up by age, largest tumor basal diameter, tumor thickness, ciliary body involved, optic disc involvement, and vitreous hemorrhage. IPW = 1/Pr in short follow-up cases and IPW = 1/(1-Pr) in long follow-up cases.

Kaplan−Meier analysis and log-rank test were performed to estimate the cumulative probability of metastasis and death. Factors predictive of melanoma-related metastasis and death according to the tumor characteristics and the presence or absence of OCM were evaluated using the nested case-control design. Significant factors were identified by univariable analysis at a 5% level of significance, and these were subsequently considered for multivariable analysis using Cox proportional hazard model by forward stepwise method. A 2-sided *P* value < 0.05 was considered for statistical significance. All analyses were performed in Stata (15.0) (StataCorp, LLC).

Results

Clinical Characteristics and Outcomes of Asian Patients with UM

Of 1151 patients with UM included in this study and divided into 3 groups (I-125 plaque brachytherapy, local resection, and enucleation) according to primary treatments (Table 1), 23 (0.20%) displayed OCM. The median follow-up duration was 24 (8−46) months. The mean tumor basal diameter was 12.2 mm, and mean tumor thickness was 7.0 mm. Mean age at presentation of UM in patients with OCM was 48.2 years compared with 46.8 years in those with no OCM. The demographic features are listed in Tables 1 and 3, and Table S1 (available at www.ophthalmologyretina.org).

Kaplan−Meier estimated the 1-, 3-, 5-, 7-, and 10-year metastasis rates and metastasis-related death in 1151 patients with UM were 2.4% (95% confidence interval [CI], 1.6−3.7), 9.5% (95% CI, 7.3−12.2), 15.5% (95% CI, 12.3−19.5), 21.4% (95% CI, 16.5−27.4), 24.5% (95% CI, 17.6−33.6), 0.7% (95% CI, 0.3−1.5), 4.3% (95% CI, 2.9−6.3), 7.5% (95% CI, 5.3−10.7), 11.9% (95% CI, 8.1−17.2), and 11.9% (95% CI, 8.1−17.2), respectively (Fig 2A and B). The IPW estimated the 1-, 3-, 5-, 7-, and 10-year metastasis rates and metastasis-related death were 3.8% (95% CI, 3.0−4.8), 13.5% (95% CI, 11.7−15.5), 19.6% (95% CI, 16.2−21.2), 21.8% (95% CI, 18.0−23.9), 23.6% (95% CI, 19.1−29.0) (Fig S2A, available at www.ophthalmologyretina.org), 0.9% (95% CI, 0.5−1.4), 7.2% (95% CI, 5.9−8.8), 10.0% (95% CI, 8.2−12.2), 10.7% (95% CI, 8.8−13.0), and 10.7% (95% CI, 8.8−13.0), respectively (Fig S2B, available at www.ophthalmologyretina.org).

Univariable Cox proportional hazards regression analysis revealed that the largest tumor basal diameter, ciliary body involvement, and tumor thickness increased the risk of metastasis and metastasis-related death. In multivariable analysis, only the largest tumor basal diameter increased the risk of both metastasis and metastasis-related death (hazard ratio [HR], 1.20; 95% CI, 1.09−1.32; HR, 1.17; 95% CI, 1.01−1.35) (Table 2).

Nested Case-Control Study Determining the Influence of OCM on UM Prognosis

A total of 23 patients with UM were diagnosed with OCM. The clinical features in eyes with UM and OCM are listed in Table 4. The median follow-up duration was 53.0 (33−67) months. The tissue sites of OCM included the conjunctiva (78%), sclera (74%), eyelid (70%), face (26%), forehead (2.2%), iris (0.87%), choroid (0.87%), and auricle (0.4%). The mean largest tumor basal

Table 3. Demographic Features of Patients with UM with OCM

No	Sex	Age at Diagnosis, yrs	Eye	BCVA before Treatment	Site of Melanocytosis	UM Quadrantic Location	Ciliary Body Involved	Retinal Detachment	Extrascleral Extension	Largest Basal Diameter (mm)	Tumor Thickness (mm)	Treatment	Follow-up Duration (Mos)	Metastasis (Time to Metastasis)	Vital Status (Time to Death)
1	M	55	R	0.6	Face, forehead, auricle, eyelid, conjunctiva, sclera, choroid	Inferior	No	No	No	12.2	6.8	PRT	56	No	Alive
2	F	40	R	0.4	Face, forehead, eyelid, conjunctiva, sclera	Superior temporal	No	Yes	No	6.9	1.9	PRT, TTT	66	No	Alive
3	M	53	L	0.6	Face, conjunctiva	Temporal	No	Yes	No	12.5	9.8	PRT	47	Liver (44)	Dead (47)
4	F	48	L	0.2	Conjunctiva	Superior temporal	No	Yes	No	7.8	8.2	PRT, enucleation	54	No	Alive
5	M	59	R	0.2	Sclera, iris, choroid	Superior nasal	No	Yes	No	8.8	8.5	PRT	31	No	Alive
6	M	57	L	0.3	Eyelid, conjunctiva, sclera	Inferior	No	Yes	No	14.4	9.5	PRT	68	No	Alive
7	M	62	L	0.01	Eyelid, conjunctiva, sclera	Nasal	Yes	Yes	No	12.9	12.3	Local resection	33	No	Alive
8	F	36	R	0.1	Sclera	Inferior Temporal	No	Yes	No	11.0	8.9	PRT	24	No	Alive
9	M	43	L	0.6	Sclera	Inferior Temporal	No	Yes	No	15.1	15.3	Enucleation	31	No	Alive
10	F	47	R	0.05	Eyelid, conjunctiva	Nasal	No	No	No	10.2	10.1	PRT	146	Liver (23)	Alive
11	F	24	R	0.1	Eyelid, conjunctiva, sclera	Superior temporal	Yes	No	Yes	13.2	16.0	PRT, enucleation	146	No	Alive
12	M	38	L	0.2	Eyelid, conjunctiva	Inferior Nasal	No	No	No	10.1	9.9	PRT	144	No	Alive
13	M	45	L	0.4	Eyelid, conjunctiva	Superior temporal	No	No	No	9.8	9.6	PRT	140	No	Alive
14	M	67	R	0.05	Face, forehead, eyelid, conjunctiva, sclera	Temporal	No	No	No	12.5	11.5	PRT	53	No	Alive
15	M	50	R	0.1	Eyelid, conjunctiva	Temporal	No	No	No	11.5	12.6	PRT	67	No	Alive
16	F	37	L	0.5	Eyelid, sclera	Superior	No	Yes	No	20.7	6.8	PRT, enucleation	57	No	Alive
17	F	51	L	CF	Conjunctiva, sclera	Macula	No	No	No	10.3	3.8	PRT	54	No	Alive
18	M	57	R	0.1	Face, forehead, eyelid, conjunctiva, sclera	Nasal	No	Yes	No	9.7	9.1	PRT	53	No	Alive
19	F	30	R	0.8	Eyelid, conjunctiva, sclera	Temporal	Yes	Yes	No	17.0	10.0	PRT	48	No	Alive
20	F	39	R	0.05	Face, forehead, eyelid, conjunctiva, sclera	Superior	No	Yes	No	8.6	7.6	PRT	39	No	Alive
21	M	62	L	0.1	Eyelid, conjunctiva, sclera	Superior temporal	No	Yes	No	9.2	2.5	TTT	38	No	Alive
22	F	50	L	0.3	Sclera, iris	Nasal	No	Yes	No	10.1	9.3	PRT	33	No	Alive
23	M	59	L	0.1	Eyelid, conjunctiva, sclera	Temporal	No	No	No	10.2	10.1	PRT	9	No	Alive

BCVA = best-corrected visual acuity; L = left; OCM = oculocutaneous melanosis; PRT = plaque radiotherapy (I-125 plaque brachytherapy); R = right; TTT = Transpupillary thermotherapy (Thermotherapy); UM = uveal melanoma.

diameter and mean tumor thickness were 11.3 mm and 9.5 mm, respectively. To fully eliminate potential confounding of tumor size and location, we performed a nested case-control study and matched 56 control patients by age, tumor thickness, largest tumor basal diameter, and ciliary body involvement. As shown in Table 4, patients with and without OCM were comparable in age, sex, and tumor size. Log-rank test shown prognosis was similar between patients with and without OCM, indicating that OCM did not increase the risk of UM metastasis (Fig 3A and B).

Discussion

Oculocutaneous melanosis is a rare condition, but it might indicate a poor predictive metastatic factor for UM in the White population. In a general population of 13 150 patients, Gonder et al[11] found melanocytosis in 1 in 6915 Black patients (0.014%) and 2 of 5251 White patients (0.038%). Velazquez and Jones[15] noted 15 of 1210 eyes enucleated for UM had melanocytosis, giving a prevalence of 1.24% in their series. In our cohort, we noted clinical evidence of melanocytosis in 23 of 1151 patients (0.20%) with UM. Previous evidence indicated that the lifetime estimates for a White patient with OCM to develop UM was 1/400 to 1/200.[17,18] Shields and colleagues[23] found that the only statistical difference involved patient race/ethnicity and that non-White individuals represented 7% of those with UM and melanocytosis compared with 2% of those without melanocytosis.

Shields and colleagues[23] estimated patients with UM associated with oculo(dermal) melanocytosis had double the risk for metastasis compared with those without melanocytosis. The location of melanocytosis was also important because the HR for melanoma-related metastasis was 2.6 for choroidal melanocytosis, 2.8 for iris melanocytosis, and 1.9 for scleral melanocytosis. On Kaplan–Meier analysis, the 10-year rate of metastasis was 48% for those with melanocytosis compared with 21.3% for those without melanocytosis. When specifically focusing on those eyes with melanoma and melanocytosis, factors for metastasis included greater tumor thickness and presence of subretinal fluid.[23]

Oculocutaneous melanosis may have a malignant transformation to UM.[27,28] Chromosomal abnormalities in UM cells, especially monosomy of chromosome 3, have been shown to be a major determinant of metastasis from UM.[29] In a study of 1059 patients with UM sampled for status of chromosomes 3, 6, and 8, OCM was found to be associated with high-risk mutation profile (chromosome 3 loss and 8q gain).[8] A recent study reported that germline FAM111B and DSC2 mutations were found among 3 cases of OCM with UM.[30] In the current era, molecular testing is the gold standard when looking at risk stratification for UM metastasis. Many centers now use monosomy 3 or GEP Class 2 as a significant risk factor for developing metastases. In most countries and regions, molecular pathology/gene analysis could not be widely used in clinical practice because of the cost and technical considerations. In this study, the main primary therapy for all patients with UM in our series is plaque brachytherapy (n = 929). The molecular pathology analysis is unavailable and may not be available for earlier patients.

In this study, we summarized clinical characteristics and prognosis information of 1151 Asian patients with UM and found that the 10-year UM metastasis and metastasis-related death rates were 24.5% (17.6%–33.6%) and 11.9% (8.1%–17.2%), respectively. In multivariable analysis, only the largest tumor diameter increased the risk of both metastasis and metastasis-related death. Neither tumor thickness nor ciliary body involvement was associated with prognosis. After matching for age, sex, and tumor size, the log-rank test showed OCM did not increase the risk of UM metastasis.

Table 4. Clinical Features in Eyes with UM and OCM in the Nested Case-Control Study

	UM with OCM (n = 23)	UM without OCM (n = 56)	*P* Value
Age, yrs	48.2 (11.1)	48.8 (8.8)	0.6
Female (%)	10 (43.5%)	29 (51.8%)	0.5
BCVA			
<0.1	5	18	0.68
0.1–0.24	9	17	
0.25–0.49	4	6	
≥0.5	5	13	
Largest tumor basal (mm)	11.3 [9.7–12.9]	10.8 [9.6–12.2]	0.95
Tumor thickness (mm)	9.5 [7.6–10.1]	9.1 [7.6–10.1]	0.62
Retinal detachment	14 (60.9%)	46 (82.1%)	0.04
Ciliary body involved	3 (13%)	6 (10.7%)	0.79
Primary treatment			
Plaque radiotherapy	20	56	
Enucleation	1	0	
Local resection	1	0	
Thermotherapy	1	0	
UM metastasis	2	2	
Metastasis-related death	1	2	

BCVA = best-corrected visual acuity; OCM = oculocutaneous melanosis; UM = uveal melanoma.

To our knowledge, this is the largest UM cohort in the Asian population. We confirmed again that there were clinical characteristic discrepancies between the Asian and White patients with UM. Compared with the White patients with UM, the patients with UM in our study were younger at diagnosis (aged ∼45 years vs. ∼58 years).[23] There were no gender differences, larger in tumor size (mean [median] 12.2 mm vs. 11.0 mm),[23] exudative retinal detachment was common (>75%), and they had a relatively good prognosis. Another significant factor for metastatic disease was the cell type of UM. Although previous studies have shown that women with UM have a better prognosis than men,[5,31] our study did not show this. Male sex did not increase the risk of metastasis and death (Table 2). The 10-year UM metastasis rate was 24.5%, which was slightly lower than in previous studies in White patients and East Asian patients.[5,32] Further genetic evidence revealed that similar genetic alterations were found between the Asian and White patients with UM.[24] More important, we found that the largest tumor basal diameter, rather than

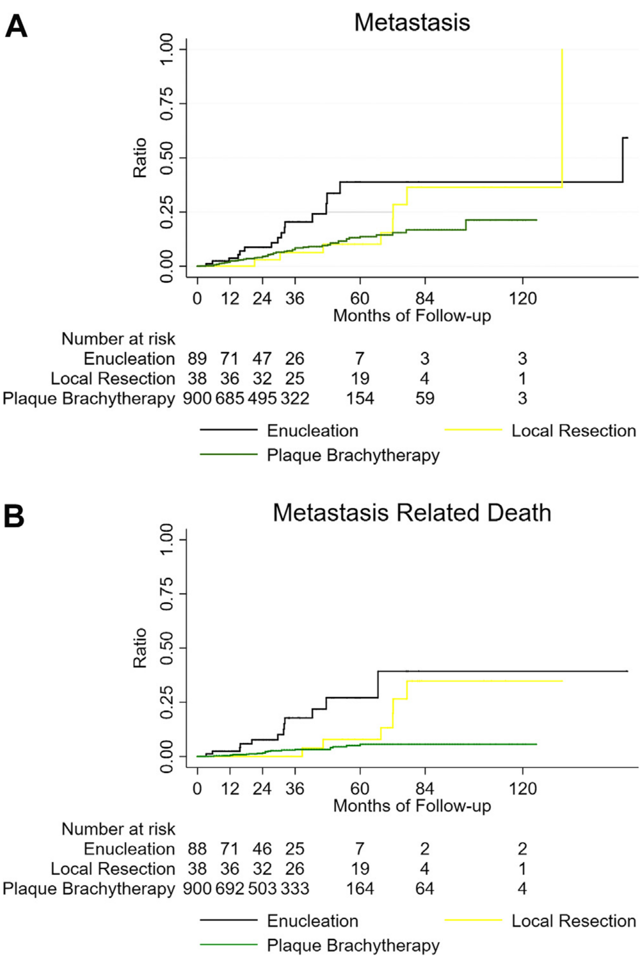

Figure 2. Survival analysis of all patients. Kaplan-Meier estimated the 1, 3, 5, 7, 10-year metastasis rate and metastasis related death (95% CI) in 1151 UM patients were 2.4% (1.6–3.7%), 9.5% (7.3–12.2%), 15.5% (12.3–19.5%), 21.4% (16.5–27.4%), 24.5% (17.6–33.6%); and 0.7% (0.3–1.5%), 4.3% (2.9–6.3%), 7.5% (5.3–10.7%),11.9% (8.1–17.2%), 11.9% (8.1–17.2%), respectively **(A, B)**.

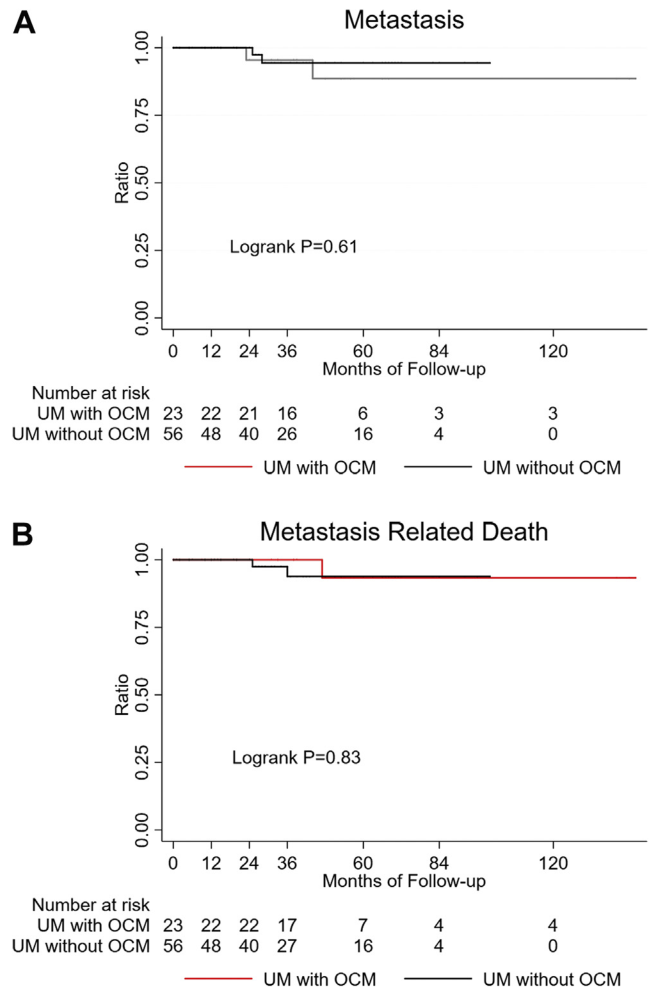

Figure 3. Survival analysis of UM patients combined with OCM and without OCM. Log-Rank test shown prognosis was similar between patients with and without OCM, indicating OCM did not increase the risk of UM metastasis **(A, B)**.

tumor thickness, ciliary body involvement, or subretinal fluid, was associated with metastasis and metastasis-related death in both univariable and multivariable analyses. These findings were partially inconsistent with previous studies,[23] which indicated that American Joint Committee on Cancer classification[33] may need to be modified when applied to Asian patients with UM.

Furthermore, we explored the relationship of OCM in the survival of Asian patients to patients with UM in this analysis. The melanocytosis as a factor did not reach significance in the outcome of melanoma-related metastasis or death. By performing a nested case-control study, Kaplan–Meier survival analysis showed that UM combined with OCM did not increase the risk of melanoma-related metastasis death. Overall, patients with UM and melanocytosis had a similar risk for metastasis compared with those without melanocytosis (log-rank test, $P < 0.68$).

This finding was inconsistent with previous studies which found that the 10-year rate of metastasis was 48% for those with melanocytosis compared with 21.3% for those without melanocytosis.[23] This discrepancy may due to racial differences. Indeed, OCM was found to be associated with a high-risk mutation profile (chromosome 3 loss and 8q gain) among White patients with UM.[8] A recent study reported that germline FAM111B and DSC2 mutations were found among 3 cases of Chinese patients with OCM and UM.[8] However, genetic alteration underlying the Asian population with OCM was still largely unknown. Considering that OCM was more common among Asians,[13] further studies were needed to evaluate its influence on UM onset among the Asian population.

However, in our analysis, we found the prominent risk for metastasis from UM was the largest tumor basal diameter and occasionally had loss to follow-up during the different lengths of follow-up. Furthermore, we found those lost to follow-up tended to be those with the largest tumor basal diameters and ciliary body involvement; therefore, the actual risk of metastasis and metastasis-related death might be underestimated, as well as the potential biases that might occur because of those differences. To estimate and eliminate the extend of selection bias, we used IPW to mitigate selection bias due to loss to follow-up. As a result, the 10-year metastasis rate and metastasis-related death rate were nearly unchanged, whereas the magnitude of risk for metastasis or death increased (Table 2). For example, ciliary body involvement is a well-known risk factor for UM metastasis. However, in the unweighted results, we failed to detect any significant association between ciliary body involvement and prognosis in multivariable analysis because those who had ciliary body involvement were likely to be lost to follow-up. After IPW weighting, ciliary body involvement increased the risk of metastasis and death (2.71 [1.82−4.02] $P < 0.001$; 2.12 [1.24−3.60] $P = 0.006$). These results show that IPW partially eliminates the select bias.

Study Limitations

First, the retrospective nature of the study and single-center research representing the normal China area may induce potential selection bias; however, all patients were examined by an experienced ophthalmologist (W.W.) over the 15 years of data collection; thus, subjects occasionally lost to follow-up may reduce bias. Second, we found those lost to follow-up tended to be those with the largest tumor basal diameters and ciliary body involvement, and patients of low socioeconomic status may receive enucleation at local hospitals. Therefore, the actual risk of metastasis and metastasis-related death might be systematically underestimated. The median follow-up time of patients with OCM was 53 (33−67) months, which showed that patients with OCM generally reached the time of end point events, and only 2 cases of metastases showed that OCM did not increase the risk of metastasis. In addition, to estimate and eliminate the extend of selection bias, we used the IPW method to mitigate selection bias due to loss to follow-up. Third, there was a lack of pathologic diagnosis for the majority of patients (plaque brachytherapy was performed), and the samples of cell type of UM were relatively small. Fourth, although we conducted a nested case-control study, the power of this sub-study to detect meaningful differences between those with and without OCM was limited by the small numbers. Considering the differences in clinical characteristics between White patients with UM and Asian patients with UM, the importance of OCM as a factor for metastasis may be different. Last, there was a lack of genetic testing analysis. Although we have estimated OCM with the metastatic rate in Asian patients with UM, there may be unknown factors that predispose these patients to the disease, such as the underlying genetic constitution of the melanocytosis. Horgan et al[34] documented chromosome 3 monosomy in UM and its absence in melanocytosis in a single case. A similar genetic alteration has been described in Asia.[29,30] Future study of the genetic analysis of OCM should be revealing to its relationship with UM.

Conclusions

Compared with White patients with UM, Asian patients with UM had different clinical characteristics, and the prominent risk for metastasis and metastasis-related death was the largest tumor basal diameter. Asian patients with UM combined with OCM had a similar risk for metastasis compared with those with no OCM.

Footnotes and Disclosures

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Author Contributions:

Conception and design: Zhou, Liu, Wei

Data collection: Zhou, Zhang, Wei

Analysis and interpretation: Zhou, Zhang, Wei

Obtained funding: N/A

Overall responsibility: Zhou, Liu, Wei

Abbreviations and Acronyms:

CI = confidence interval; **HR** = hazard ratio; **IPW** = inverse probability-of-censoring weighted; **OCM** = oculocutaneous melanosis; **UM** = uveal melanoma.

Keywords:
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